

DICOM Series Viewer - Version 2.0

1. Document Control

VERSION	STATUS	OWNER	DATE	CHANGES
1.0	Draft	Project Team	2023-05-10	Initial Version
2.0	Final	Project Team	2025-08-10	Added features, updated details

2. Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements for a **DICOM Series Viewer**. This software will allow medical professionals to load, view, analyze, and export DICOM image series efficiently. Key features include:

- DICOM file loading and rendering
- Slice navigation
- Window width/level (WW/WL) adjustment
- Distance measurement
- Cine playback
- Image export (PNG, MP4)

3. Scope

In-Scope:

- Loading and viewing of a single DICOM series.
- Image display with proper rescaling (using RescaleSlope and RescaleIntercept).
- Interactive windowing adjustments (WW/WL).
- Slice navigation via mouse, keyboard, and sliders.
- Metadata display (patient info, slice-specific data).
- Distance measurement between two points on the image.
- Cine playback for DICOM series.
- Export of DICOM slice or cine sequence to PNG or MP4.

Out-of-Scope:

- Multi-series comparison.
- Advanced image segmentation or annotations beyond basic distance measurement.
- DICOM network communication (C-STORE, C-FIND).
- Multi-planar reconstructions (MPR) or 3D volume rendering.

4. Definitions and Acronyms

- **DICOM**: Digital Imaging and Communications in Medicine.
- **WW/WL**: Window Width/Window Level (used to adjust image contrast and

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brightness).

- **ROI:** Region of Interest, used for distance measurement.
- **PHI:** Protected Health Information, subject to data privacy laws.
- **Cine Playback:** Sequence playback of images (slice animation).
- **PixelSpacing:** Physical distance between the centers of adjacent pixels.

5. References

- **Python 3.x** (version 3.9 or higher).
- Libraries: pydicom, numpy, matplotlib, imageio.
- DICOM standards: Ensure compatibility with DICOM metadata and pixel data (RescaleSlope, PixelSpacing).

6. Overall Description

6.1 Users

- **Primary Users:** Medical professionals and students (radiologists, clinicians).
- **Secondary Users:** Developers and testers.

6.2 System Context

- **Platform:** Windows 10 (with cross-platform support for Linux/macOS).
- The application does not rely on any network communication for its core functionality and operates solely on locally stored DICOM files.

6.3 High-Level Capabilities

- **DICOM Series Loading:** Import and render image series from DICOM files.
- **Image Calibration:** Apply rescale slope/intercept to pixel data for accurate intensity representation.
- **Interactive Windowing:** Adjust window width and level (WW/WL).
- **Slice Navigation:** Navigate through slices using sliders, keyboard, or mouse.
- **Metadata Display:** Show relevant metadata (patient information, pixel spacing, etc.).
- **Distance Measurement:** Measure distances on the image using pixel spacing.
- **Cine Playback:** Playback DICOM series as a sequence of images.
- **Export:** Save images as PNG or export cine sequence to MP4.

7. Functional Requirements

FR-01: Load DICOM Series

- **Description:** Load DICOM files from a user-selected directory and display them in sequence.
- **Input:** Directory path containing DICOM files.
- **Output:** Display the DICOM images in the viewer.

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FR-02: Image Calibration

- **Description:** Apply RescaleSlope and RescaleIntercept to pixel data to display the correct intensity values.
- **Input:** DICOM pixel data and metadata.
- **Output:** Calibrated image ready for display.

FR-03: Windowing (WW/WL)

- **Description:** Provide interactive slides to adjust the window width and window level of the image for contrast and brightness.
- **Input:** Slider values for WW/WL.
- **Output:** Adjusted image contrast and brightness.

FR-04: Slice Navigation

- **Description:** Allow users to navigate between slices using a slider, keyboard, or mouse.
- **Input:** Slider or keyboard/mouse input.
- **Output:** The slice display changes accordingly.

FR-05: Metadata Display

- **Description:** Display metadata for each DICOM slice, such as patient name, ID, study date, modality, and pixel spacing.
- **Input:** Metadata from the loaded DICOM files.
- **Output:** Displayed metadata panel.

FR-06: Distance Measurement

- **Description:** Allow users to select two points on an image and calculate the distance between them using pixel spacing.
- **Input:** Coordinates of two points on the image.
- **Output:** Display distance between the selected points.

FR-07: Cine Playback

- **Description:** Provide a play/pause button to view the DICOM series in sequence.
- **Input:** Play/pause button input.
- **Output:** Animated sequence of slices displayed.

FR-08: Export

- **Description:** Allow users to export the current image as a PNG or the entire cine sequence as an MP4 file.
- **Input:** Export format selection (PNG/MP4).
- **Output:** PNG image or MP4 cine sequence file.

FR-09: Command-Line Interface (CLI)

- **Description:** Accept the directory path as a command-line argument to load the DICOM series.
- **Input:** Directory path as a command-line argument.
- **Output:** Loaded DICOM series.

8. External Interface Requirements

8.1 User Interface

- **Image Viewer:** Main window with a canvas for displaying images and controls for interaction (sliders, buttons).
- **Metadata Panel:** Panel displaying metadata for the currently selected slice.
- **Navigation Controls:** Sliders for WW/WL adjustments, slice navigation, and playback controls.

8.2 Hardware Interfaces

- **Input Devices:** Mouse and keyboard for navigation and interaction.
- **Display:** Minimum resolution of 1280x800 for optimal user experience.

8.3 Software Interfaces

- **Operating System:** Windows 10 (with support for Linux/macOS).
- **Dependencies:** pydicom, numpy, matplotlib, imageio.

8.4 File Interfaces

- **Input:** DICOM files (extensions: .dcm, .dicom).
 - **Output:** PNG image and MP4 video.
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9. Non-Functional Requirements

NFR-01: Performance

- The system should load and display a DICOM series with up to 1,000 slices in under 5 seconds.
- The system should respond to WW/WL and slice navigation adjustments in under 100 milliseconds.

NFR-02: Usability

- The user interface should be intuitive and easy to use.
- Controls for windowing, slice navigation, and playback should be responsive and clearly labeled.

NFR-03: Reliability

- The application should handle incomplete or corrupted DICOM files gracefully, without crashing.
- Metadata should be displayed even if some information is missing.

NFR-04: Portability

- The application should be compatible with Windows 10, Linux, and macOS (cross-platform).

NFR-05: Security and Privacy

- No network communication is required.
- The system does not process Protected Health Information (PHI).

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NFR-06: Maintainability

- The code should be modular, with clear separation of concerns between different components.
- The system should follow industry-standard coding practices and be well-documented.

NFR-07: Accessibility

- The system should provide keyboard shortcuts for all interactive controls (windowing, slice navigation, and playback).

10. Constraints

- **UI Framework:** Matplotlib-based UI may have backend-specific limitations.
- **Performance:** Loading and rendering large series (e.g., 2,000+ slices) may require higher-end hardware.
- **External Libraries:** Dependencies like pydicom and numpy should be maintained and up to date to ensure compatibility with the latest DICOM formats.